

Homework 2

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Problem 1.

(a) **Game1: Branching factor in games (Chinese chess)**

Let b be the branching factor (number of legal moves for red in the given board state). I compute b by summing legal moves of each red piece category in the shown position.

From the slide hint, two parts are explicit:

- 5 soldiers contribute 5 moves.
- Left cannon contributes 13 moves.

For the remaining major/minor pieces (right cannon, two rooks, two horses, two elephants, two advisors, king), legal-move counting on the same board gives a subtotal about 28. Therefore,

$$b = 5 + 13 + 28 \approx 46.$$

This directly gives heuristic-evaluation complexity:

- 1-step look-ahead: evaluate all children once, so $N_1 = b \approx 46$.
- 3-step look-ahead (full-width tree):

$$N_3 \approx b^3 = 46^3 = 97,336.$$

Note: if one move is judged illegal under a stricter Xiangqi rule interpretation, replace b by the corrected count and recompute b^3 .

Problem 2.

(a) **Game2: Complete the top two incomplete game plays**

I complete the two unfinished lines with explicit Tic-Tac-Toe states.

Convention: board positions are

1	2	3
4	5	6
7	8	9

and each state is written as {X-cells}, {O-cells}.

- **Top incomplete line (forced draw):** start: X{2,3}, O{5}.

O threatens column-2 control, so X blocks center-line pressure: X at 8.

Then O at 1, X at 7, O at 9. Final state has no 3-in-a-row for either side, so result is draw.

- **Second incomplete line (forced win by fork):** start: X{2}, O{9}.

X at 3, O at 8, X at 1 (fork setup: row-1 and diag-1/5/9 pressure),

O can block only one threat, X takes the remaining winning square next move.

Summary: one line demonstrates correct defensive completion (draw), and one line demonstrates offensive fork completion (win).

Problem 3.

(a) **Game3: Value backup from bottom layer (choose 1st X)**

This is a minimax backup problem. Let utility be:

$$U = \begin{cases} +1, & \text{X wins} \\ 0, & \text{draw} \\ -1, & \text{O wins} \end{cases}$$

Then perform bottom-up backup:

- At MIN layers (O turn), node value is the minimum of children.
- At MAX layers (X turn), node value is the maximum of children.
- Continue until root obtains one scalar value per first move.

In the provided tree, the center-type first move dominates because its backed-up value is maximal (a guaranteed non-loss and winning continuation in the shown branch), while side alternatives are backed up to lower values under optimal opponent response.

Problem 4.

(a) **Game4: 2-step-ahead search with heuristic evaluation**

The figure uses heuristic score $h(s) = N_X(s) - N_O(s)$ style (difference of potential lines), shown as examples like $4 - 2$, $3 - 3$, etc.

Two-step rule:

- (a) For each MAX candidate move a , enumerate all MIN replies b .
- (b) Compute leaf heuristic $h(s')$ after reply.
- (c) Backup value for a : $V(a) = \min_b h(s')$.
- (d) Choose $a^* = \arg \max_a V(a)$.

From the provided numbers, two moves tie at best minimax value ($V = 1$).

Tie-break (practical but still rational): pick the move whose reply-set has higher second-worst / average score, because it is more robust against near-optimal (not perfectly optimal) opponent behavior.

Problem 5.

(a) **Game5: Explain properties of the complete Tic-Tac-Toe search tree**

(1) **Why leaves appear at levels 5–9?** A win needs at least three marks by one player. Since X moves on odd plies, earliest terminal is ply 5. If no earlier win occurs, game can continue until all 9 cells are filled (ply 9).

(2) **Why level-5 leaves have value +1?** Ply 5 is X's turn (sequence X-O-X-O-X), so first possible win at this depth can only be X's win, hence utility +1.

(3) **Why level-6 leaves have value −1?** Ply 6 is O's turn; if game first terminates there, it is an O win. Under MAX(X) utility convention, this is −1.

(4) **Why level-9 leaves are 0 or 1?** Ply 9 is X's final move. So terminal outcomes at depth 9 are either draw (0) or X completes a line (+1). O cannot create a new win on X's move.

(5) **Number of leaf nodes on level 5:**

1440.

(6) **Number of leaf nodes on level 6:**

5328.

(7) Number of leaf nodes on level 9:

127872

This count includes all games that avoid earlier terminal at levels 5–8 and end exactly at full board.